

Backend and AI engineer building software since December 2022. I connect product interactions to reliable systems: voice agents, scoped tools, integrations, realtime collaboration, payments, and operations. Core backend and AI work at Blitzit; previously co-founded and led engineering at MaddyCustom.

01 · Experience

Core Backend Developer & AI Engineer

Blitzit

Nov 2025 - Present

AI PRODUCTIVITY PLATFORM · UK, REMOTE · TYPESCRIPT · FASTIFY · MONGODB · REDIS · BULLMQ

- Built core backend and AI systems for Blitzit 3.0, spanning task APIs, auth, recurrence, notifications, collaboration, billing, and resumable migration from the legacy platform.
- Built a provider SDK for 12 integrations and an MCP surface with 70+ tools, OAuth 2.1/PKCE, typed schemas, and shared execution for in-app agents and external clients. Integrated bidirectional sync, webhooks, retry decisions, and provider-specific recovery.
- Designed a reversible change journal for human and AI mutations, with content checks to prevent undo from silently replacing newer edits. Built shared-list permissions and realtime delivery; optimized change-stream connection use.
- Built Blitzy's voice architecture: realtime speech delegates actions to the existing reasoning agent, preserving scoped tools and context. Implemented transcript handling, interruption, and coordinated audio output. Backend merged; complete frontend experience in review.
- Trained a custom "Hey Blitzit" wake-word model in Colab and integrated local ONNX inference, candidate transcription verification, microphone handoff, and a state-driven orb with audio-reactive animation and offscreen rendering controls.

Co-Founder & Lead Full-Stack Developer & AI Engineer

MaddyCustom

Dec 2022 - Feb 2026

CUSTOM VEHICLE COMMERCE · NEXT.JS · MONGODB · RAZORPAY · PAYU · SHIPROCKET · AWS · CLERK

- Led engineering across the custom storefront and admin platform for a business reaching 100K+ monthly users and approximately \$62.4K USD annual revenue. Connected catalogue variants, custom offers, checkout, inventory, production, and fulfillment.
- Built health-aware payment provider selection and fallback across Razorpay and PayU, server-side verification, guarded post-payment handling, split-order totals, and shipping lifecycle integration.
- Built the team's operations tools: production-template downloads, design search, order management, packaging, tax exports, review moderation, and department access.
- Built first-party funnel and customer-journey analytics, UTM attribution, period comparisons, repeat-buyer analysis, and server-side conversion tracking. Built a multi-agent shopping assistant with typed catalogue and order tools, product cards, follow-up pagination, policy retrieval, and human handoff.

02 · Selected Projects

- Spyll PERSONAL PROJECT · 2023 ONWARDS · College social product with verified access, anonymous posts and polls, Socket.IO chat, Redis-backed random matching, moderation, and queued notifications.
- Avana FREELANCE · SOLE DEVELOPER · 2025 · AI investment research assistant for Bali real estate: orchestrator-style multi-agent chat (OpenAI Agents SDK), realtime WebRTC voice, document-backed answers over vector search, Xendit subscriptions, and an admin panel for documents, news scraping/approval, and knowledge ops. Two codebases shipped solo.
- AutoRemov FREELANCE · BACKEND ENGINEER · 2026 · Backend for an image background-removal product: presigned uploads, durable pg-boss job pipeline, credit and billing flows, Razorpay billing, MySQL→Postgres/Prisma migration, and integration tests around payments and processing.
- Dailicle FOUNDER · 2025 · Weekly deep-research publication: reading surface, archive, subscriber emails, audio support, and publishing workflow (Next.js, MongoDB, AWS S3, CloudFront).

03 · Skills

PRIMARY	TypeScript, Node.js, Fastify, MongoDB, PostgreSQL, Redis, BullMQ, Zod
AI / AGENTS	MCP, OpenAI Agents SDK, realtime voice, tool execution, RAG, ONNX Runtime, wake-word training
DISTRIBUTED SYSTEMS	OAuth 2.1/PKCE, webhooks, queues, event-driven architecture, idempotency, REST, JSON-RPC, WebSockets, SSE, WebRTC, realtime sync
ALSO	React, Next.js, Flutter, Python, Docker, AWS (S3, CloudFront), Cloudflare, Prisma, Playwright, Razorpay, Xendit, admin tooling and analytics

04 · Education